

The empirical recurrence for OEIS A283631: recovering a conjecture that is not written down

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Abstract

OEIS A283631 records “Empirical recurrence of order 63”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 118$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 63, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 63 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A283631 is “Number of nX5 0..1 arrays with no 1 equal to more than one of its horizontal, diagonal and antidiagonal neighbors, with the exception of exactly two elements.”. It has offset 1 and begins

2, 88, 2875, 61592, 1185010, 20051838, 317384829, 4754748994, ...

The entry states:

Empirical recurrence of order 63 (see link above)

As of the “Last modified” line on the live entry (Mar 12 2017 14:29:49, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1286$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 118$ of the 1286, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 118$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 236$ exact terms of the model returns order 63 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{63} c_i a(n-i),$$

c_1	39	c_{17}	4356153054	c_{33}	51834868142	c_{49}	339777597
c_2	−552	c_{18}	16332102468	c_{34}	−50023744050	c_{50}	−58318518
c_3	2980	c_{19}	−15126039774	c_{35}	−77916224118	c_{51}	−51028564
c_4	3486	c_{20}	−51494786118	c_{36}	13344332148	c_{52}	13390608
c_5	−111252	c_{21}	21197056676	c_{37}	70709062062	c_{53}	5333166
c_6	396213	c_{22}	717694518	c_{38}	−3704760792	c_{54}	−1926571
c_7	774531	c_{23}	149512166382	c_{39}	−38875953030	c_{55}	−366315
c_8	−8599569	c_{24}	−117587927754	c_{40}	−7018967478	c_{56}	186201
c_9	12522165	c_{25}	−11484068436	c_{41}	20973480186	c_{57}	13397
c_{10}	65569992	c_{26}	−135626369874	c_{42}	7517339780	c_{58}	−11982
c_{11}	−236774538	c_{27}	156341799016	c_{43}	−11821460034	c_{59}	72
c_{12}	−88919894	c_{28}	81058977690	c_{44}	−2849937882	c_{60}	474
c_{13}	1495690656	c_{29}	−35786606898	c_{45}	5195205284	c_{61}	−30
c_{14}	−1185336111	c_{30}	−111123244242	c_{46}	344087106	c_{62}	−9
c_{15}	−3956394035	c_{31}	−61454465208	c_{47}	−1591814646	c_{63}	1
c_{16}	3016766766	c_{32}	134579552280	c_{48}	108968403		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 63, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $63 + 4$ terms removed returns order 63 as well, so $\deg q_{\min} = 63 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $63 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 63 that this sequence satisfies — and that family turns out to have exactly one member.

References

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